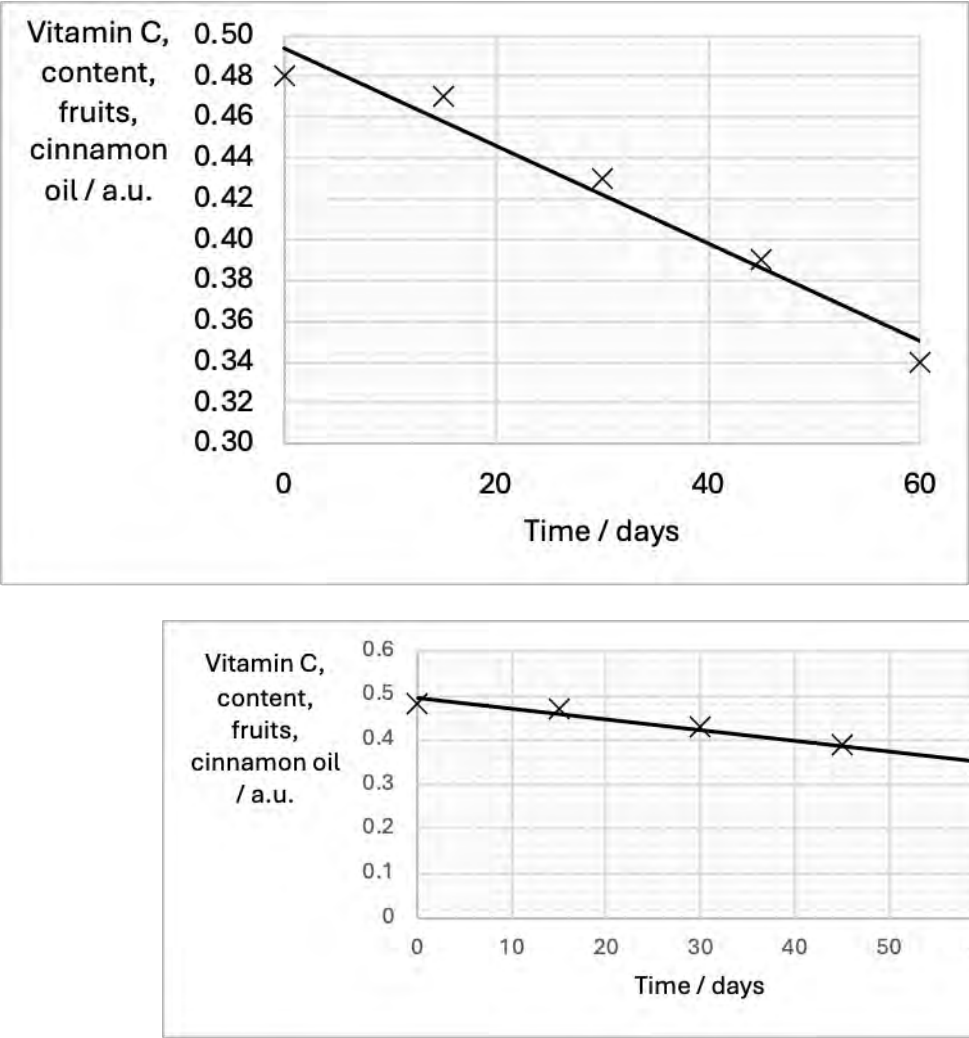


Question number	Answer	Additional guidance	Mark
3(a)(i)	- percentage decrease of fruit coated with chitosan calculated (1) - difference calculated correctly and given to given to 2 significant figures (1)	34.69387755, or any correct rounding (34.6938776, 34.693878, 34.69388, 34.6939, 34.694, 34.69, 34.7) for the calculation ACCEPT 35 for one mark only e.g. $43.75 - 34.69 = 9.06$ $= 9.1$ to 2 sig figs Incorrect rounding at any intermediate stage can only have 1 mark.	(2)

Question number	Answer	Additional guidance	Mark
3(a)(ii)	An answer that includes the following points: • A-axes correct (x- time, y - Vit Conc,) (1) • L-both axes labelled with suitable title and units (x, Time / days, y, Vitamin C content of fruits coated with cinnamon oil / a.u. (1) • points plotted correctly on linear scale (1) • appropriate line of best fit (1)	 The top graph shows data points at approximately (0, 0.49), (15, 0.47), (30, 0.43), (45, 0.39), and (60, 0.34). The bottom graph shows data points at approximately (0, 0.48), (15, 0.46), (30, 0.42), (45, 0.39), and (60, 0.34). Both graphs include a line of best fit.	(4)

Question number	Answer	Additional guidance	Mark
3(a)(iii)	A description that includes the following points: • repeat {for all 3 groups / experiment} (1) • (calculate) mean and standard deviation (1) • plot (mean and) {standard deviation / error bars} (on a graph) (1)	IGNORE formula ACCEPT SD ACCEPT find ACCEPT range bars ACCEPT present in a table described e.g. with SD column	(3)

Question number	Answer	Additional guidance	Mark
3(a)(iv)	A description that includes three of the following points: • describe method to standardise coating method (1) • repeat the investigation under the same conditions as before (1) • use of DCPIP (1) • description of titration (1)	e.g. dip in it for standard time / do as done in original investigation / same concentrations as in original investigation ACCEPT e.g. temperature, length of time, 60 days, size of fruit, type of fruit IGNORE colour change e.g. titrate dropwise / using burette / pipette	(3)

Question number	Answer	Additional guidance	Mark
3(b)(i)	A description that includes the following points: • take readings at the (bottom of the) meniscus (1) • hold the pipette at eye level when taking the readings (1)	ACCEPT reverse argument ACCEPT avoid parallax error ACCEPT perpendicular	(2)

Question number	Answer	Additional guidance	Mark
3(b)(ii)	• 1×10^5 / 1.0×10^5	ACCEPT $1 \cdot 10^5$ $1.0 \cdot 10^5$	(1)

Question number	Answer	Additional guidance	Mark
3(b)(iii)	An explanation that includes two of the following points: • {aseptic technique / description of aseptic technique} to avoid contamination of {cultures with other bacteria / ourselves with bacteria (which could be pathogenic)} (1) • {leave the lid loose / only partially seal lid} (of Petri dish onto base) so that {conditions are aerobic / anaerobic bacteria (which could be pathogenic) do not grow} (1) • growing bacteria at temperatures {slightly below 37°C / no higher than 35°C / body temperature} to decrease growth of pathogenic bacteria (1)	ACCEPT harmful, disease causing for pathogenic IGNORE irritant, allergy IGNORE to kill microbes ACCEPT infection ACCEPT taped vertically ACCEPT forming pathogenic	(2)

Question number	Answer	Additional guidance	Mark
3(b)(iv)	An answer that includes the following points: • different types of agar used as bacteria and fungi have different nutritional requirements (1) • higher temperature for bacteria because the {enzymes / metabolism} of bacteria have a higher optimum (temperature than the enzymes of fungi) (1) • fungi cultured for longer as they have a slower {growth / reproductive} rate (than bacteria) (1)	ACCEPT food ACCEPT reverse argument ACCEPT enzyme / metabolism optima different ACCEPT reverse argument ACCEPT growth rates different ACCEPT fungi take more time to grow	(3)

Question number	Answer	Additional guidance	Mark
3(b)(v)	An answer that includes the following points: • uses {a range / more than one species} (of microorganism) / bacteria and fungi} so conclusion is (more) representative • uses only four microbes so conclusion is not representative (1) • no {data given for uncoated fruit / control} so do not know if either coating has an (antimicrobial) effect (1) • no {error bars / SDs / repeats} so cannot judge if the {differences are statistically significant / there is no measure of variability (1) • measures diameter, direct area measure would be better <i>because</i> zone not always circular (1) • no indication of concentration of coating substance so may not be {representative / optimal} (1)	ACCEPT the differences are small so cannot judge if the differences are statistically significant	(4)